

The Calhoun Forecasts

In business plans dated 2001 and 2013, Stephen Calhoun described connected home services and outdoor interactive learning experiences. Many elements later appeared in commercial products, while others remained partial or niche. Each passage is quoted *in his own words* and compared with later developments. Observations of trends already underway when the plans were written are left out. Dates mark illustrative milestones, not a uniform measure of widespread adoption.

July 30, 2001 The CES plan June 2013 The GenZPlay plan June 2025 SASKI begins Twelve years apart, each time

NEW WIRING

Combined parts that already existed.

These tended to arrive sooner.

NEW INVENTION

Needed something nobody had built yet. These generally took longer.

MARKET CALL

A forecast about people, markets or timing rather than a product.

The CES plan

WRITTEN JULY 30, 2001 · PAGE NUMBERS REFER TO THE ORIGINAL PLAN

Quotes keep the original 2001 spelling and punctuation. Dates mark later milestones in the US, such as launches, releases or adoption figures.

Strong hits 7 ENTRIES

WHAT YOU WROTE IN 2001	WHAT IT BECAME	YOUR EXACT WORDS	LATER MILESTONES	TYPE
Microsoft putting Word and Excel on the Internet for a monthly usage fee , with no downloads and updates available immediately p . 3	The plan recognized a model Microsoft was already exploring and extended it to everyday customers. Before 2010, Office over the Internet was mainly for businesses, which rented it from hosting providers. Microsoft's 2001 subscription test covered installed software in only three countries and was dropped. Free browser based Office Web Apps arrived in 2010, Office 365 launched for businesses in 2011, and the version for regular customers, Office 365 Home Premium at \$9.99 a month, arrived in 2013. It is now Microsoft 365.	<i>"Microsoft has recently moved down the road towards putting applications (i.e. Word, Excel) on the Internet to be accessed, instead of installed inside of everybody's PC... What better way to sell somebody a product then directly thru the Internet with a monthly usage fee? Users would not have to download the latest updates or wait days for their new software to arrive in the mail. The latest and greatest business and recreational software will be available immediately for our use."</i>	June 2010 (Office Web Apps), June 2011 (Office 365 for business), January 2013 (Office 365 Home Premium for consumers). Renamed Microsoft 365 in April 2020. 9 to 12 years later	NEW WIRING
A book downloaded wirelessly onto a lightweight screen the size of a hardback novel, which opens to the last page read by you, your spouse or others , on a folding screen p . 7	Strong anticipation, with individual features realized across several products rather than one device. The Kindle delivered wireless downloads, Whispersync syncs the last page read across a person's devices, and family sharing lets a household share books. Folding screens arrived in 2019, on phones rather than books.	<i>"You pick up your new book (downloaded wirelessly) on a folding screen in the shape of a hardback novel (but lighter in weight) which would open to the last page read by yourself, your spouse or others."</i>	November 2007 (Kindle), February 2009 (Whispersync), 2014 (Kindle Family Library), September 2019 (Galaxy Fold). 6 to 18 years later	NEW INVENTION
Choosing your own video or music service online, and saving on your cable bill by getting entertainment through the Internet connection p . 9	YouTube, Netflix streaming, Spotify, and the whole cord cutting movement. Netflix already rented DVDs by mail in 2001. The forecast was entertainment delivered through the Internet connection itself, replacing part of the cable bill.	<i>"Online video rental and pizza ordering are amongst the various business aspects that we can bring to our customers. The customized appearance of the customers website would allow them to choose what grocery stores do they like, which video or music service do they prefer... How can they save money on their cable bill by receiving entertainment thru the Internet connection?"</i>	December 2005 (YouTube launch), January 2007 (Netflix streaming), July 2011 (Spotify in the US). 4 to 10 years later	NEW WIRING

WHAT YOU WROTE IN 2001	WHAT IT BECAME	YOUR EXACT WORDS	LATER MILESTONES	TYPE
Posting family pictures to a family website and keeping a shared family calendar pp . 3 , 9	Facebook, shared photo albums, and shared Google and Apple calendars.	<i>"How about checking all of your email and spending a few minutes checking out the latest pictures posted to the family web site from your family members." / "They can post pictures of the family and update the family calendar and contacts."</i>	February 2004 (Facebook at Harvard), April 2006 (Google Calendar), September 2006 (Facebook opens to everyone). 3 to 5 years later	NEW WIRING
Your cellphone turning up the A/C and lights as you get close to home p . 3	Location based automations in Nest, SmartThings, Google Home and Apple Home. In 2001 most US phones couldn't report their location, and there were no phone app stores.	<i>"Your cellphone will turn up the A/C and turn on the lights as you get close to."</i>	2011 (Nest thermostat), 2014 to 2016 (phone location automations in Nest, SmartThings and Apple Home). 10 to 15 years later	NEW INVENTION
A GPS route that bypasses traffic congestion , with spoken weather and traffic alerts as you get in the car p . 8	Live traffic routing in Waze and Google Maps, read aloud through CarPlay and Android Auto. In 2001, consumer GPS units in the US had no live traffic data and couldn't reroute around congestion.	<i>"Upon entering your car and the closing of your door, the garage door is lifted, seatbelt reminder and weather alerts are spoken to you along with traffic advisories and a new GPS programmed route, which will bypass any traffic congestion on your way to the office for the day."</i>	2007 (Google Maps live traffic), 2009 (Waze), 2014 (CarPlay), 2015 (Android Auto). 6 to 14 years later	NEW INVENTION
Ordering pizza, a lawn service and a pool cleaner in a few clicks , and switching contractors with a click instead of a confrontation pp . 3 , 9	Angi (formerly Angie's List and HomeAdvisor), Thumbtack, TaskRabbit and DoorDash. Your core business idea became a large category.	<i>"How about clicking a few check boxes and ordering pizza, a new lawn-mowing service and a pool cleaner in just a few minutes." / "The customer can change the contractor they use with the click of a button instead of a face-to-face confrontation, which so many people feel uncomfortable with."</i>	June 2004 (ServiceMagic operating nationally), mid to late 2000s (Angie's List goes national), 2008 (TaskRabbit and Thumbtack), 2012 (ServiceMagic becomes HomeAdvisor), 2013 (DoorDash). 3 to 12 years later	NEW WIRING

Niche, partial, late or missed 12 ENTRIES

WHAT YOU WROTE IN 2001	WHAT IT BECAME	YOUR EXACT WORDS	LATER MILESTONES	TYPE	VERDICT
Kitchen touch panel with news, calendar, email, security cameras, child monitors and alerts for water heater, plumbing, HVAC and propane p . 7	Echo Show and Nest Hub with Ring or Nest camera feeds, plus leak sensors and smart propane monitors. The pieces all exist, but few homes have the full integrated setup.	<i>"Kitchen touch panel displays CNN, which appears in one corner and your daily appointments in another along with email and voicemail alerts for all phones. Other corners can be security cameras, child monitors, home alert devices (hot water system, plumbing system, hvac system, propane levels), and grocery list."</i>	June 2017 (Echo Show), October 2018 (Google Home Hub). 16 to 17 years later	NEW WIRING	Came true, partly The full integrated version is still niche.
One tap "goodnight" and morning sequences for lights, security, thermostat, alarm music and coffee maker pp . 3 , 7	Alexa and Google routines with smart plugs and thermostats.	<i>"You load your coffee maker the night before, hop into bed, hit the touch panel to turn all house lights off, except your bedroom, arm the security system and turn on your alarm clock including the setting of your morning wake-up music, the thermostat gets set to the appropriate level and your home/cell phones are set to mute." / "Your alarm clock will start the coffee maker."</i>	2017 (Alexa Routines), 2018 (Google Assistant Routines). 16 to 17 years later	NEW WIRING	Came true
Smart refrigerator that builds a grocery list, sends it to the store for delivery and flags spoiled food p . 3	Later smart refrigerators implemented several elements, such as internal cameras, shopping lists and grocery ordering. Expiration tracking still relies on dates the owner enters.	<i>"Refrigerators will be equipped with barcode scanners and will make your grocery list for you and send it to the local store for delivery and they will also alert you to food that is spoiling or outdated."</i>	May 2016 (Samsung Family Hub on sale in the US). 15 years later	NEW INVENTION	Partly realized LG showed a \$20,000 Internet fridge in 2000 that almost nobody bought. No product has delivered the fully automated system the plan described.
Door locks and security resets automatically when you leave, and the garage door opens as you get in the car to leave p . 8	Smart locks and app controlled garage openers like myQ.	<i>"As the door shuts, it is locked and security system is reset to proper level for wife's awakening, kitchen lights and coffee maker are shut off. Upon entering your car and the closing of your door, the garage door is lifted"</i>	2011 (Chamberlain myQ), 2013 (August Smart Lock). 10 to 12 years later	NEW WIRING	Came true

WHAT YOU WROTE IN 2001	WHAT IT BECAME	YOUR EXACT WORDS	LATER MILESTONES	TYPE	VERDICT
Remote car start with automatic cabin temperature, automatic shutoff, and carbon monoxide protection in the garage p . 8	App based remote start and climate preconditioning, standard on Teslas and many other cars.	<i>"Your car is started automatically (shuts off in 5 minutes if you have not entered vehicle or if Carbon Monoxide emissions reach unsafe level within garage due to ventilation failure) and car temperature level automatically set depending upon outside temperature."</i>	2010 (OnStar RemoteLink app), 2012 (Tesla Model S preconditioning). 9 to 11 years later	NEW INVENTION	Partly realized Remote start and automatic climate came true. The carbon monoxide safety shutoff is still not standard.
Alert that the school bus is coming down your street p . 7	School bus tracking apps used by many districts.	<i>"A reminder flashes on the screen to make sure that the children have headed out to the curbside as the school bus has just been sensed starting down your street."</i>	Mid 2010s (apps such as Here Comes the Bus). About 13 years later	NEW WIRING	Came true
Automatic pet feeding and watering and a pet door that unlocks p . 7	Smart feeders and microchip pet doors.	<i>"the dog and cat bowls are auto filled with food and water" / "Pet door unlocks to allow the dog and cat out."</i>	Late 2000s (microchip pet doors), mid 2010s (app controlled feeders). 7 to 14 years later	NEW WIRING	Came true, niche
Bills paid automatically on the day they are due, or all at once with a few clicks pp . 3 , 7	Autopay became common for utilities and other recurring bills.	<i>"Appropriate funds are automatically being transferred to accounts for Utility bills that are due that day." / "How about paying all of those bills at once or at specified times during the month with just a few clicks."</i>	Mid 2000s (mainstream bank bill pay and utility autopay). About 4 to 5 years later	NEW WIRING	Came true Online bill pay existed in 2001 but few households used it. Paying automatically on the due date became common in the mid 2000s.
Trash can that scans barcodes onto your shopping list, and reminders on a screen in the bathroom p . 7	Barcode scanning trash cans have been built. Mirrors with built in displays exist but remain rare.	<i>"Empty can of shaving cream thrown into wastebasket where it is bar code scanned upon entry and added to computerized shopping list." / "Short list of daily reminders/ appointments is on screen in bathroom or shower for daily planning and clothing decisions while showering and shaving."</i>	2016 (GeniCan barcode trash scanner). Display mirrors remain niche products. 15 years later	NEW INVENTION	Partly realized, novelties

WHAT YOU WROTE IN 2001	WHAT IT BECAME	YOUR EXACT WORDS	LATER MILESTONES	TYPE	VERDICT
75% of computer users with high speed access within 2 years p . 3	Home broadband did become the norm.	<i>"In the next 2 years, 75% of computer-users in the cities and rural communities will have high-speed Internet access."</i>	Roughly 2008 , when about three quarters of US Internet users had home broadband (Pew). The plan measured computer users, so this comparison is approximate. Several years later than forecast	MARKET CALL	Late Right direction, but slower than the two years forecast.
"Contractor brokering has no competition" p . 17	The category became large, led by Angie's List and ServiceMagic (later HomeAdvisor, now Angi).	<i>"The major service that CES is offering, contractor brokering, has no competition to speak of, as there is currently no other company offering this type of service."</i>	1995 (Angie's List, Midwest) and December 1998 (ServiceMagic, Denver) had already started. ServiceMagic was operating nationally by June 2004 .	MARKET CALL	Overstated Competitors existed, though both were still small in 2001. The integrated household service CES proposed may still have been distinctive.
Ordering all utilities once with a single form when you move p . 3	Moving concierge services handle part of it.	<i>"Imagine sitting down at the computer the week before your moving into your new home and ordering all of your utilities (i.e. Electricity, Phone, Gas, Rubbish, Cable, Internet, Security, etc) online with just entering your Name, address, phone numbers and SSN just once...no more waiting 20 minutes on the phone for each service."</i>	2010 (Updater founded), still only partial today.	NEW WIRING	Missed so far Most people still set up utilities one provider at a time.

The GenZPlay plan

WRITTEN JUNE 2013 · SECTION NAMES REFER TO THE ORIGINAL PLAN

GenZPlay built outdoor, interactive fitness and learning apps for kids. The plan reports that a demo version of its first app, Big Cat Race, had been released for iOS and Android and had reached 10,000 downloads. The other apps and hardware described below were proposed or in development. Forecasts the plan quotes from other sources are left out; only Stephen's own predictions are included.

Strong hits 7 ENTRIES

WHAT YOU WROTE IN 2013	WHAT IT BECAME	YOUR EXACT WORDS	LATER MILESTONES	TYPE
Interactive stories played outdoors in your backyard or park, using real trees and structures, bringing the virtual world out of the home Executive Summary, Conclusion	Pokémon Go got millions of kids and families walking outside to play. Niantic's Ingress was an invite only beta in June 2013 and didn't launch publicly until December 2013.	<i>"Imagine an interactive story with many possible outcomes that takes place in your own backyard, or park, using the trees, plants and structures within." / "These methods will bring the virtual world out of the home, which is where children have found their stimulation as of today."</i>	July 2016 (Pokémon Go). 3 years later	NEW WIRING
A device on a child's wrist, hip or backpack that becomes second nature, with a motivator built in Executive Summary, Industry Findings	Activity trackers already existed, as the plan notes (Zamzee and pedometers). What followed were mainstream kids' wearables, activity rings, and Apple Watches for kids who don't have their own iPhone.	<i>"We want to show the children nature, exploration, mystery and fitness with an electronic device on their wrist, hip or in their backpack. The electronic device should become second nature to all of us" / "An app is an inexpensive way to have a motivator built into the devices we use on a daily basis."</i>	April 2015 (Apple Watch activity rings), 2016 (Garmin vivofit jr.), 2018 (Fitbit Ace), September 2020 (Apple Watch Family Setup). 2 to 7 years later	NEW WIRING
Doctors recognizing the healing power of apps and recommending them to families. The plan's opening also names attention deficit disorder Problem & Opportunity, Marketing	A useful later milestone rather than a direct match. EndeavorRx became the first game based treatment the FDA authorized, prescribed to improve attention function in children ages 8 to 12 with ADHD.	<i>"Doctors, health teachers and psychotherapists are realizing the healing impact that the new apps are bringing to the world." / "GenZPlay has spoken with many doctors who would be willing to promote the GenZPlay apps as a way of giving extra advice and help for families."</i>	June 2020 (EndeavorRx). 7 years later	NEW INVENTION

WHAT YOU WROTE IN 2013	WHAT IT BECAME	YOUR EXACT WORDS	LATER MILESTONES	TYPE
Body sensors and eHealth as "the next dot.com boom" Problem & Opportunity	Digital health venture funding hit a record \$29.1 billion, then fell sharply in 2022 and 2023, much like the original dot com cycle. The plan called the boom, not the bust.	<i>"with the current acceleration of body sensor devices and eHealth products, we are seeing the next dot.com boom."</i>	2021 (record funding year). 8 years later	MARKET CALL
Teachers turning into "entertainment creators" Problem & Opportunity, Conclusion	Game based classroom tools such as Kahoot, Gimkit and Blooket.	<i>"As the educational system is now transforming from babbling and confused teachers to entertainment creators, their goal is always the same, and currently is not how to improve communication with their students but how to entertain their students with what they want to learn."</i>	Fall 2013 (Kahoot public launch), 2017 to 2018 (Gimkit and Blooket). Months to 5 years later	NEW WIRING
4K televisions as the next big trend for custom installers Custom Integration	4K became the standard TV. The first consumer 4K sets went on sale in late 2012, so the plan caught the transition just as installers were starting to adapt to it.	<i>"Currently, the trend is for security and video systems, and the next trend will most likely be the new 4K televisions"</i>	2014 (Netflix 4K streaming), 2018 (over 100 million 4K sets sold worldwide in one year, close to half of all TVs). 1 to 5 years later	MARKET CALL
A Bluetooth tracker on an object , with an app that guides you to where it is Custom Integration	A related implementation. Tile, then Apple AirTag with Precision Finding, handle the finding part. The adventure with app generated clues remains a separate idea. Tile opened its crowdfunding the same month the plan was written.	<i>"Allow the parents to mount a BT tracker to a prize and place it somewhere on their property. The app will setup an adventure and use clues to guide them towards its location."</i>	2014 (Tile ships), April 2021 (AirTag). 1 to 8 years later	NEW WIRING

Niche, partial, late or missed 10 ENTRIES

WHAT YOU WROTE IN 2013	WHAT IT BECAME	YOUR EXACT WORDS	LATER MILESTONES	TYPE	VERDICT
Electronic Tag & Chase with Bluetooth LED pods: tap one and the next lights up, and the app times the course Custom Integration	BlazePod: Bluetooth pods you tap as they light up, run from an app that times your reactions.	<i>"Attach a single or multiple LED's to a BT tracker and mount them to trees, rocks, etc around the yard. The app will create a random courses and time them as they complete it. Children can also tap on the LED/BT tracker and then the next one would light up to extend the course, or follow the lights like a game of interactive TAG."</i>	March 2017 (BlazePod Kickstarter). About 4 years later	NEW INVENTION	Came true, closely A close match to the interaction the plan described, down to tapping a pod to light the next one.
Augmented reality Frisbee golf built on Bluetooth hardware Executive Summary	TechDisc, a Bluetooth sensor disc that measures each throw and simulates its flight. The UDisc scoring app already existed. The augmented reality visual layer the plan described has not arrived.	<i>"Bluetooth hardware for Virtual 'Augmented Reality' Frisbee Golf"</i>	2012 (UDisc app), early 2020s (TechDisc). About 8 years later	NEW INVENTION	Partly came true The Bluetooth hardware came true. The augmented reality part is still unfulfilled.
Game style nature apps for kids exploring outdoors Competition	The plan notes that plant and bird identification apps already existed. Its idea was turning outdoor exploration into a game for kids, which Seek by iNaturalist later did with badges and challenges. Merlin added bird identification by sound.	<i>"This is the beauty of our world outside of the house, but all gaming is currently focused on fantasy worlds explored from the couch. The outdoor world is the best world there is to explore, so lets create apps for that."</i>	2018 (Seek), 2021 (Merlin Sound ID). 5 to 8 years later	NEW WIRING	Came true, niche Popular with families, small next to mainstream gaming.
A portable encyclopedia with storylines that adapt to how each child learns Executive Summary	The plan proposed adaptive stories and learning content tied to physical activity. Adaptive tutoring already existed, dating back to the 1990s. Generative AI tools like ChatGPT and Khanmigo later expanded what personalization can do.	<i>"allow our children to hold a portable encyclopedia of fun and learning. We also need to build storylines and content that will adapt to the children's learning preferences and excite them to be active."</i>	November 2022 (ChatGPT), March 2023 (Khanmigo). About 10 years later	NEW INVENTION	Partly realized Personalized learning has grown, but adaptive stories tied to physical activity remain rare.

WHAT YOU WROTE IN 2013	WHAT IT BECAME	YOUR EXACT WORDS	LATER MILESTONES	TYPE	VERDICT
A middle school education through massive multiplayer learning apps Problem & Opportunity	Game based learning like Minecraft Education entered classrooms. Separately, virtual schools and the 2020 switch to remote learning moved coursework online, but not through multiplayer learning games.	<i>"Will a 'Middle School' degree be available, from the use of Massive Multiplayer Educational apps, before you graduate on to an actual High School? Many menial courses/classes can be bypassed or completed via an app model."</i>	November 2016 (Minecraft Education Edition), 2020 (widespread remote school). 3 to 7 years later	MARKET CALL	Partly came true A full education delivered through multiplayer learning games has not emerged.
Technology to help with children's diabetes Industry Findings	Dexcom Share lets parents see a child's glucose readings on their own phones.	<i>"If we combine technology with kinesiology programs and other health related issues, such as children's diabetes, we can teach, entertain and coach our kids to be the person they want to be."</i>	October 2014 (Dexcom Share cleared by the FDA). 1 year later	NEW INVENTION	Came true, partly On the monitoring side, not as teaching and coaching games.
Smart backyard features: LED landscapelighting, yard speakers, solar charging, playhouses with electronic whiteboards Custom Integration	App controlled outdoor lighting is common. Outdoor speakers were already standard work for installers.	<i>"LED's can double as landscape accent lighting" / "Speakers in the yard for music and gameplay" / "Solar charging station for electronics" / "Integrated playhouses with electronic white boards"</i>	2018 (Philips Hue Outdoor). 5 years later	NEW WIRING	Mixed Smart outdoor lighting came true. Electronic whiteboard playhouses never caught on.
About a year left before the industry peaked, with the technology within 12 months Advantage	The big moment for outdoor games was Pokémon Go in 2016. BlazePod's pods followed in 2017.	<i>"This window of opportunity is just opening and we have about a year to perfect and market our products before the industry pinnacle." / "Creating this technology is easy, as the hardware and software systems are already in place or are within 12 months of fruition."</i>	2016 to 2017. 3 to 4 years later	MARKET CALL	Late Right direction, but 3 to 4 years instead of 1.
Patents and copyright wouldn't hold back apps Business Model	App developers faced patent suits and copycat crackdowns.	<i>"The only difference is, in the past, you had to create something absolutely unique due to patents and copyright infringements. The app market is nothing like this."</i>	Already happening: 2011 (Lodsys sues app developers), August 2012 (Apple v. Samsung verdict).	MARKET CALL	Missed This was already untrue when the plan was written.

WHAT YOU WROTE IN 2013	WHAT IT BECAME	YOUR EXACT WORDS	LATER MILESTONES	TYPE	VERDICT
Parents would grow more comfortable with kids' technology as Gen Y became parents Problem & Opportunity	It went the other way: The Anxious Generation, school phone bans in many US states, and Australia's social media ban for children under 16.	<i>"The Baby-Boomer parents, who didn't quite understand how the world changed so fast with the introduction of TV's and computers, instigated the recent apprehension of technology use with our children... We now know that technology is just a more entertaining looking glass into a world of education and understanding, so we need to take back time and give children quality access to technology."</i>	2024 (The Anxious Generation, state phone bans), December 2025 (Australia's ban takes effect).	MARKET CALL	Mixed The plan expected more comfort with technology while warning against passive, absorbing use. Worry grew, but the optimism was qualified, so this is not a clear cut miss.

What the two plans show

Specific experiences, not vague trends.

The plans describe concrete things a person would do, like lights coming on as you approach home or tapping a light pod to extend a race. That specificity is what makes them checkable.

A twelve year rhythm.

The plans were written in 2001 and 2013, and Stephen started SASKI in 2025. Each came a few years after a new platform arrived: the web, the smartphone, and AI. Both plans also ask the same question: why technology isn't giving families more of their time back.

New wiring tended to arrive sooner.

In this set, ideas that combined parts already on the market often showed up within a few years. Ideas that needed something new, like a folding screen or phone location triggers, generally took much longer.

Direction ahead of timing.

The market calls were mostly right about where things were heading and often early about when. The timing forecasts in both plans were optimistic.

SOURCES FOR DATES AND FACTS

- Angie's List, Encyclopedia of Indianapolis
- How Angie's List Started (Powderkeg interview with Angie Hicks)
- HomeAdvisor (Wikipedia)
- IAC Acquires ServiceMagic (Feld Thoughts)
- Seek app lets kids enjoy nature (Forbes)
- BlazePod review (Gadgets & Wearables)
- UDisc (Wikipedia)
- From trial and error to TechDisc (UDisc blog)
- Annual 4K TV sales to surpass 100 million (FlatpanelsHD)
- BlazePod Kickstarter campaign
- ServiceMagic partnership announcement, June 2004 (NACHI)
- Samsung Family Hub US availability (Samsung)
- Facebook opens to everyone, 2006 (Meta)
- Carnegie Mellon cognitive tutors, 1998 (CMU)
- Office Web Apps history (Microsoft)
- Microsoft Releases Office 365 Home Premium (Microsoft)
- Microsoft dumps Office subscription trial (Computerworld)
- Digital health funding reaches record \$29.1B in 2021 (Healthcare Dive)

Other launch dates are drawn from widely reported public records. Compiled September 2026.